

QFT1 Formulas

July 17, 2026

1 Basics of QFT

1.1 Path integrals

Definition:

$$Z = \int \mathcal{D}\phi \exp \left(i \int d^4x \mathcal{L}[\phi, \partial_\mu \phi] \right) = \int \mathcal{D}\phi e^{iS[\phi]} \quad (1)$$

Where

$$\int \mathcal{D}x = \lim_{N \rightarrow \infty} \int \prod_{i=0}^N dx_i \quad (2)$$

To get to that form from a correlation function:

- Start from some correlation function between value at two times t_0 and t_f .
- Discretize time into N intervals of $\Delta t = (t_f - t_0)/N$, such that $t_N = t_f$.
- Insert a complete set of states $\int d\phi_i |\phi_i\rangle \langle \phi_i| = \mathbf{1}$ at each time step t_i .
- Evaluate the infinitesimal transition amplitudes $\langle \phi_{i+1} | e^{-iH\Delta t} | \phi_i \rangle$.
- Take the continuum limit $N \rightarrow \infty, \Delta t \rightarrow 0$, transforming the product of integrals into the functional measure $\mathcal{D}\phi$.

1.2 Correlation functions

$$\langle \Omega | T \{ \phi(x_1) \dots \phi(x_n) \} | \Omega \rangle = \frac{\int \mathcal{D}\phi \phi(x_1) \dots \phi(x_n) e^{iS[\phi]}}{\int \mathcal{D}\phi e^{iS[\phi]}} \quad (3)$$

Time ordering operator

Places operators with later times to the left of operators with earlier times.
For bosons:

$$T\{\phi(x)\phi(y)\} = \begin{cases} \phi(x)\phi(y)\theta(x^0 - y^0) + \phi(y)\phi(x)\theta(y^0 - x^0) & \text{for bosons} \\ \phi(x)\phi(y)\theta(x^0 - y^0) - \phi(y)\phi(x)\theta(y^0 - x^0) & \text{for fermions} \end{cases} \quad (4)$$

Normal ordering operator

Places all creation operators ($a^\dagger$) to the left of all annihilation operators (a). Its vacuum expectation value is exactly zero.

Vacuum state definitions

- $|0\rangle$: The vacuum state of the *free* theory ($H_0|0\rangle = 0$).
- $|\Omega\rangle$: The true, interacting vacuum state of the full Hamiltonian H .

Wick's theorem

$$T\{\phi_1 \dots \phi_n\} = N\{\phi_1 \dots \phi_n\} + \text{all possible contractions} \quad (5)$$

Inside $\langle 0 | \dots | 0 \rangle$, only the fully contracted terms survive.

1.3 Source

A classical external field $J(x)$ added to the Lagrangian to probe the system: $\mathcal{L} \rightarrow \mathcal{L} + J(x)\phi(x)$.

$$Z[J] = \int \mathcal{D}\phi e^{i \int d^4x (\mathcal{L} + J\phi)} \quad (6)$$

Correlation functions are generated by taking functional derivatives of $Z[J]$ with respect to J :

$$\langle \Omega | T\{\phi(x_1) \dots \phi(x_n)\} | \Omega \rangle = \frac{(-i)^n}{Z[0]} \frac{\delta^n Z[J]}{\delta J(x_1) \dots \delta J(x_n)} \Big|_{J=0} \quad (7)$$

1.4 Scattering

The transition matrix can be split into unity + a scattering matrix

$$S = \mathbf{1} + iT \quad (8)$$

Scattering matrix (with scattering amplitude):

$$\langle f | iT | i \rangle = (2\pi)^4 \delta^{(4)}(p_f - p_i) i\mathcal{M} \quad (9)$$

Cross section:

$$d\sigma = \frac{1}{2E_A 2E_B |v_A - v_B|} |\mathcal{M}|^2 d\Pi_{\text{LIPS}} \quad (10)$$

where

$$d\Pi_{\text{LIPS}} = (2\pi)^4 \delta^{(4)}(p_A + p_B - \sum p_f) \prod_f \frac{d^3 p_f}{(2\pi)^3 2E_f} \quad (11)$$

Effective potential (in the non relativistic limit with distinguishable particles):

$$\langle p' | iT | p \rangle = -i\tilde{V}(\vec{p}' - \vec{p}) (2\pi) \delta(E_{p'} - E_p) \quad (12)$$

Decay rate

For a single particle of mass M decaying into a final state f :

$$d\Gamma = \frac{1}{2M} |\mathcal{M}|^2 d\Pi_{\text{LIPS}} \quad (13)$$

1.5 Lippmann-Schwinger equation

Describes scattering states in quantum mechanics.

$$|\phi_{\pm}\rangle = |\phi_0\rangle + \frac{1}{E_0 - H_0 \pm i\epsilon} V |\phi_{\pm}\rangle \quad (14)$$

Where $|\phi_0\rangle$ is the eigenstate of the free Hamiltonian H_0 with eigenvalue E_0 , V is the interaction potential ($H = H_0 + V$), and the $\pm i\epsilon$ defines the boundary conditions (incoming/outgoing spherical waves).

1.6 Schwinger-Dyson equations

Steps:

- Utilize the invariance of the path integral measure under an infinitesimal field shift $\phi(x) \rightarrow \phi(x) + \epsilon(x)$.
- This implies the functional integral of a total derivative is zero: $\int \mathcal{D}\phi \frac{\delta}{\delta\phi(x)} \left(e^{iS[\phi]} \dots \right) = 0$.
- Expand the derivative to relate correlation functions of different orders.

Example - the ϕ^4 equations:

- Start with the identity: $\int \mathcal{D}\phi \frac{\delta}{\delta\phi(x)} \left(\phi(y) e^{iS[\phi]} \right) = 0$.
- Apply the functional derivative: $\int \mathcal{D}\phi \left(\delta^{(4)}(x-y) + i\phi(y) \frac{\delta S}{\delta\phi(x)} \right) e^{iS[\phi]} = 0$.
- Substitute the classical equation of motion $\frac{\delta S}{\delta\phi(x)} = -(\Box_x + m^2)\phi(x) - \frac{\lambda}{3!}\phi^3(x)$.
- Divide by $Z[0]$ to yield the time-ordered expectation value relation:

$$(\Box_x + m^2) \langle \Omega | T \{ \phi(x) \phi(y) \} | \Omega \rangle + \frac{\lambda}{3!} \langle \Omega | T \{ \phi(x)^3 \phi(y) \} | \Omega \rangle = -i\delta^{(4)}(x-y) \quad (15)$$

1.7 Ward-Takahashi identities

Steps:

- Assume a continuous symmetry in the Lagrangian (e.g., local gauge invariance).
- Perform a local change of variables in the path integral corresponding to the symmetry transformation.
- Demand that the path integral remains invariant, extracting the quantum analogue of Noether's current conservation.

Example - Complex ϕ^4 / QED (Global $U(1)$):

- Perform a local transformation $\phi(x) \rightarrow e^{i\alpha(x)}\phi(x) \approx \phi(x) + i\alpha(x)\phi(x)$.
- The variation of the action is $\delta S = -\int d^4x j^\mu(x) \partial_\mu \alpha(x)$.

- The path integral measure is invariant, so $\delta Z = 0$.
- Differentiate the invariant path integral $\int \mathcal{D}\phi \mathcal{D}\phi^* e^{iS} \phi(y_1) \dots \phi^*(z_1)$ with respect to $\alpha(x)$.
- Integration by parts yields the divergence of the current $\partial_\mu j^\mu$ inside the correlator, proportional to contact terms (delta functions) at the insertion points.

$$\partial_\mu \langle T \{ j^\mu(x) \phi(y_1) \dots \phi^*(z_1) \dots \} \rangle = -i \sum_k \delta^{(4)}(x - y_k) \langle \dots \rangle + i \sum_j \delta^{(4)}(x - z_j) \langle \dots \rangle \quad (16)$$

1.8 Composite operators

Point Splitting Regularization

When evaluating operators at the same spacetime point, e.g., $j^\mu = \bar{\psi}(x) \gamma^\mu \psi(x)$, products of fields at the exact same point are singular. We separate them by a small distance ϵ :

$$\bar{\psi}(x) \gamma^\mu \psi(x) \longrightarrow \lim_{\epsilon \rightarrow 0} \bar{\psi}(x + \epsilon/2) \gamma^\mu \exp \left(-ie \int_{x-\epsilon/2}^{x+\epsilon/2} A_\nu dz^\nu \right) \psi(x - \epsilon/2) \quad (17)$$

The Wilson line exponential is required to maintain local gauge invariance.

Callan-Symanzik Equation

For an n -point function with an insertion of a composite operator $\mathcal{O}(x)$, the equation picks up the anomalous dimension of the operator $\gamma_{\mathcal{O}}$:

$$\left[\mu \frac{\partial}{\partial \mu} + \beta(\lambda) \frac{\partial}{\partial \lambda} + n\gamma(\lambda) + \gamma_{\mathcal{O}}(\lambda) \right] G^{(n, \mathcal{O})}(x_1 \dots x_n; y) = 0 \quad (18)$$

Composite operators Renormalization

Composite operators require their own renormalization. Because they share quantum numbers, different bare operators can mix under renormalization, requiring a mixing matrix Z_{ij} :

$$\mathcal{O}_i^{(0)} = \sum_j Z_{ij} \mathcal{O}_j^{(R)} \quad (19)$$

where $\mathcal{O}_i^{(0)}$ is the bare operator and $\mathcal{O}_j^{(R)}$ are the renormalized ones.

2 Regularization

2.1 Dimensional Regularization

Steps:

- Identify that the integral needs regularization
- change the dimension of the momentum integral $d_0 \rightarrow d$
- Use Feynman's trick to resolve the propegators into a single fraction
- Linearly change variables to complete the square in the denominator $l = p + \dots$
- Name the remaining part of the denominator Δ
- Remove odd powers of l
- Perform Wick's rotation $l^0 \rightarrow i l_E^0$
- Split by powers of l and use the known integrals:

$$\frac{d^d l_E}{(2\pi)^d} \frac{1}{(l_E^2 + \Delta)^n} = \frac{1}{(4\pi)^{\frac{d}{2}}} \frac{\Gamma\left(n - \frac{d}{2}\right)}{\Gamma(n)} \Delta^{\frac{d}{2} - n} \quad (20)$$

$$\frac{d^d l_E}{(2\pi)^d} \frac{l_E^2}{(l_E^2 + \Delta)^n} = \frac{d/2}{(4\pi)^{\frac{d}{2}}} \frac{\Gamma\left(n - \frac{d}{2} - 1\right)}{\Gamma(n)} \Delta^{\frac{d}{2} + 1 - n} \quad (21)$$

- Set the dimension $d = d_0 - \epsilon$ and use the approximation of Γ :

$$\Gamma(\epsilon) = \frac{1}{\epsilon} - \gamma + \mathcal{O}(\epsilon) \quad (22)$$

and of powers of Δ :

$$\Delta^\epsilon = 1 - \epsilon \ln \Delta + \mathcal{O}(\epsilon^2) \quad (23)$$

Notes:

- It preserves both Lorentz invariance and gauge invariance, making it the preferred method for gauge theories like QED and QCD.
- It struggles intrinsically with defining the γ^5 matrix since γ^5 is strictly a 4-dimensional object.

2.2 Pauli-Villars Regularization

Steps:

- Introduce fictitious, heavy particles with mass Λ to regulate high-momentum behavior by altering the propagator: $\frac{1}{k^2-m^2} \rightarrow \frac{1}{k^2-m^2} - \frac{1}{k^2-\Lambda^2}$.
- Combine the fractions and remove vanishing terms. Only converging terms should remain.
- Use Feynman's trick and the subsequent steps from dimensional regularization above.

Notes:

- Preserves local $U(1)$ gauge invariance in QED but breaks non-Abelian gauge symmetries (QCD).
- The fictitious particles violate unitarity (negative probability/ghosts), so they must be removed by taking $\Lambda \rightarrow \infty$ after renormalization.
- Note: PV regularization fails for diagrams that diverge worse than logarithmically (e.g., the $\mathcal{O}(\lambda)$ 2-point function in ϕ^4 which diverges linearly/quadratically because reducing the power of k in the denominator by just 2 is insufficient), where dimensional regularization remains effective.

3 Renormalization

3.1 Superficial Degree of Divergence

The total power of momentum in a loop integral. Definition:

$$D = d \cdot L - 2P = d + V \left(n \frac{d-2}{2} - d \right) - N \frac{d-2}{2} \quad (24)$$

Where d is the spacetime dimension, L is the number of loops, P is the number of internal propagators, V is the number of vertices, and N is the number of external legs. The coupling originates from a ϕ^n interaction. For a ϕ^4 theory in $d = 4$, this simplifies to $D = 4 - N$.

Diagram categorization

- $D > 0$: Superficially divergent (e.g., $D = 2$ is quadratic divergence).
- $D = 0$: Superficially logarithmically divergent.
- $D < 0$: Superficially convergent (though it may contain divergent sub-diagrams).

Theory categorization

- **Super-renormalizable**: Only a finite number of Feynman diagrams are divergent (e.g., coupling has positive mass dimension).
- **Renormalizable**: Only a finite number of *amplitudes* diverge; divergences appear at all loop orders but can be absorbed by a finite number of counterterms (e.g., coupling is dimensionless).
- **Non-renormalizable**: All amplitudes diverge at sufficiently high loop order; requires an infinite number of counterterms (e.g., coupling has negative mass dimension).

3.2 Field Strength Renormalization

The parameter Z describing the probability that the bare field creates a 1-particle state from the vacuum. The renormalized field is defined as $\phi_R = \frac{1}{\sqrt{Z}}\phi_0$. It is determined by the derivative of the self-energy $\Sigma(p^2)$:

$$Z = \left(1 - \frac{\partial \Sigma(p^2)}{\partial p^2} \Big|_{p^2=m^2} \right)^{-1} \quad (25)$$

Using the Källén-Lehmann spectral representation:

$$\langle \Omega | T \{ \phi(x) \phi(y) \} | \Omega \rangle = \int_0^\infty \frac{dM^2}{2\pi} \rho(M^2) \Delta_F(x-y; M^2) \quad (26)$$

The spectral density $\rho(M^2)$ contains a delta function for the 1-particle state at the physical mass m^2 . The field strength renormalization $Z = |\langle \Omega | \phi(0) | \lambda_0 \rangle|^2$ is the coefficient of this pole in momentum space:

$$\Delta(p) = \frac{iZ}{p^2 - m^2 + i\epsilon} + \int_{\sim 4m^2}^\infty \frac{dM^2}{2\pi} \rho(M^2) \frac{i}{p^2 - M^2 + i\epsilon} \quad (27)$$

3.3 LSZ reduction formula

Relates S-matrix elements to the time-ordered Green's functions of the interacting theory:

$$\int \dots e^{ip \cdot x} \langle \Omega | T \{ \phi(x_1) \dots \phi(x_n) \} | \Omega \rangle = \left(\prod_j \frac{i \sqrt{Z}}{p_j^2 - m^2} \right) \langle f | S | i \rangle \quad (28)$$

3.4 The optical theorem

$$2\text{Im}\mathcal{M}(k_A, k_B \rightarrow k_A, k_B) = \sum_f \int d\Pi_f |\mathcal{M}(k_A, k_B \rightarrow f)|^2 = 2E_{cm} p_{cm} \sigma_{\text{tot}} \quad (29)$$

A direct consequence of the unitarity of the S-matrix ($S^\dagger S = \mathbb{K}$). It relates the imaginary part of the forward scattering amplitude to the total scattering cross-section.

The cross section on the right hand side involves a sum over all possible on-shell, physical multi-particle intermediate states that the initial particles can annihilate into. It serves as a mathematical statement of the conservation of probability in scattering processes.

3.5 Renormalized perturbation theory

(This is how we actually calculate things):

- Make sure the Lagrangian has all possible terms, allowed by symmetries.
- Express the bare Lagrangian in terms of renormalized fields, masses, and couplings ($\phi_0 = Z^{1/2}\phi$, $m_0 = Z_m m$, etc.).

- Define the physical masses and couplings, and split the Lagrangian to create counterterms.
- Treat the counterterms as new interaction vertices in Feynman diagrams.
- Choose a renormalization scheme (e.g., $\overline{\text{MS}}$, $\overline{\text{MS}}$, or on-shell) to specify the renormalization conditions.
- Calculate the physical parameters based on their definitions, and use their definition to fix the counterterms.
- Write down the final expression using these counterterms.

Renormalization schemes

- **On-Shell Scheme:** Counterterms are chosen such that the renormalized mass m is exactly the physical pole mass, and the field strength renormalization yields $Z = 1$. This enforces $\Sigma(m^2) = 0$ and $\left. \frac{d\Sigma}{dp^2} \right|_{p^2=m^2} = 0$.
- **$\overline{\text{MS}}$ (Minimal Subtraction):** Counterterms are chosen to exactly cancel the $1/\epsilon$ poles arising in dimensional regularization, and nothing else.
- **$\overline{\text{MS}}$ (Modified Minimal Subtraction):** Counterterms cancel the $1/\epsilon$ pole along with the ubiquitous constants $\ln(4\pi)$ and the Euler-Mascheroni constant γ_E that always naturally accompany the pole.

3.6 The beta function

The β function encodes the running of the coupling constant λ with the energy scale μ .

$$\beta(\lambda) = \mu \frac{\partial \lambda}{\partial \mu} = \frac{\partial \lambda}{\partial \ln \mu} \quad (30)$$

The Callan–Symanzik Equation

Describes how the n -point Green's functions shift under a change in the renormalization scale μ :

$$\left[\mu \frac{\partial}{\partial \mu} + \beta(\lambda) \frac{\partial}{\partial \lambda} + n\gamma(\lambda) \right] G^{(n)}(x_1 \dots x_n; \mu, \lambda) = 0 \quad (31)$$

where $\gamma(\lambda) = \frac{1}{2} \mu \frac{\partial}{\partial \mu} \ln Z$ is the anomalous dimension of the field.

Classification of theories by beta function

- $\beta(\lambda) < 0$: The coupling decreases at high energies (Asymptotic Freedom). Characteristic of Non-Abelian gauge theories like QCD.
- $\beta(\lambda) > 0$: The coupling grows at high energies (Triviality / Landau pole). Characteristic of QED and ϕ^4 theory.
- $\beta(\lambda) = 0$: The coupling is scale invariant (Conformal Field Theory).

3.7 The Renormalization Group

Wilsonian effective action

Instead of dealing with counterterms, one integrates out high-momentum degrees of freedom (from a cutoff Λ down to a lower scale Λ').

$$Z = \int_{0 < k < \Lambda} \mathcal{D}\phi e^{iS[\phi]} = \int_{0 < k < \Lambda'} \mathcal{D}\phi e^{iS_{\text{eff}}[\phi]} \quad (32)$$

This integration generates an infinite number of new interactions in the effective action S_{eff} .

Wilsonian beta function

Tracking the coefficients (couplings) of these generated operators as the cutoff is lowered yields the Wilsonian RG flow equations. Couplings correspond to Relevant ($D > 0$), Marginal ($D = 0$), or Irrelevant ($D < 0$) operators depending on if they grow, stay constant, or decay at low energies.

$$\beta_{Wil}^I(g) = \Lambda \frac{\partial g^I}{\partial \Lambda} = \gamma_J^I g^J + \beta_{JK}^I g^J g^K \quad (33)$$

Renormalization group equation

Describes the flow of couplings in model space as a function of the threshold energy. In ϕ^n theory:

$$\begin{aligned} \frac{\partial g_n}{\partial \Lambda} = \frac{1}{2} \int \frac{d^4 p}{(2\pi)^4} \left[\sum g_{k+1}(p_1, \dots, p_k, p) \delta(p_1 + \dots + p_k + p) \frac{\partial K^{-1}(p^2)}{\partial \Lambda} \times \right. \\ \left. \times g_{n-k+1}(p_{k+1}, \dots, p_n, -p) - \frac{\partial K^{-1}}{\partial \Lambda} g_{n+2}(p_1, \dots, p_n, -p) \right] \end{aligned} \quad (34)$$

Renormalization group flow

Linearizing the Wilsonian beta function around a fixed point g^* :

$$\lambda^I = g^I - g^{*I} \quad \beta(g^*) = 0 \quad (35)$$

$$\Lambda \frac{\partial}{\partial \Lambda} \lambda^J = \gamma_J^I \lambda^J \quad (36)$$

Diagonalizing to solve the system of PDEs gives another set of couplings g'^I :

$$\Lambda \frac{\partial}{\partial \Lambda} \lambda'^I = \gamma^I \lambda'^I \quad (37)$$

- $\gamma^I > 0 \rightarrow$ Irrelevant
- $\gamma^I < 0 \rightarrow$ Relevant

4 Known Theories

Theories that we saw in class

4.1 Bare propagators by field

Scalar:

$$\Delta_F(p) = \frac{i}{p^2 - m_0^2 + i\epsilon} \quad (38)$$

Dirac Fermion:

$$S_F(p) = \frac{i(\not{p} + m_0)}{p^2 - m_0^2 + i\epsilon} \quad (39)$$

Photon:

$$D_F^{\mu\nu}(p) = \frac{-i}{p^2 + i\epsilon} \left(\eta^{\mu\nu} - (1 - \xi) \frac{p^\mu p^\nu}{p^2} \right) \quad (40)$$

Photon (Feynman gauge, $\xi = 1$):

$$D_F^{\mu\nu}(p) = \frac{-i\eta^{\mu\nu}}{p^2 + i\epsilon} \quad (41)$$

4.2 $2 \rightarrow 2$ scattering and Mandelstam variables

Mandelstam variables:

$$s = (p_1 + p_2)^2 = (p_3 + p_4)^2 \quad (42)$$

$$t = (p_1 - p_3)^2 = (p_2 - p_4)^2 \quad (43)$$

$$u = (p_1 - p_4)^2 = (p_2 - p_3)^2 \quad (44)$$

Note: $s + t + u = \sum_{i=1}^4 m_i^2$.

4.3 Real scalar ϕ^4 theory

Full Bare Lagrangian

$$\mathcal{L} = \frac{1}{2}(\partial_\mu \phi)(\partial^\mu \phi) - \frac{1}{2}m^2\phi^2 - \frac{\lambda}{4!}\phi^4 \quad (45)$$

Symmetries

Poincaré invariance, discrete $\mathbb{Z}_2$ parity ($\phi \rightarrow -\phi$).

Equations of Motion

$$(\square + m^2)\phi + \frac{\lambda}{3!}\phi^3 = 0 \quad (46)$$

Superficial Degree of Divergence

$$D = 4 - N \quad (47)$$

Where N is the number of external legs.

Feynman rules

Element	Position Space	Momentum Space
Propagator	$\Delta_F(x - y)$	$\frac{i}{p^2 - m^2 + i\epsilon}$
Vertex	$-i\lambda \int d^4x$	$-i\lambda(2\pi)^4 \delta^{(4)}(\sum p)$
External line	1	1

Counter terms and Renormalized Lagrangian

$$\mathcal{L} = \frac{1}{2}(\partial_\mu \phi_R)^2 - \frac{1}{2}m^2 \phi_R^2 - \frac{\lambda}{4!}\phi_R^4 + \frac{1}{2}\delta_Z(\partial_\mu \phi_R)^2 - \frac{1}{2}\delta_m \phi_R^2 - \frac{\delta_\lambda}{4!}\phi_R^4 \quad (48)$$

Counterterm Propagator insertion: $i(p^2 \delta_Z - \delta_m)$

Counterterm Vertex: $-i\delta_\lambda$

On-Shell Renormalization

We fix the physical mass m and field strength $Z = 1$ by demanding the propagator has a pole at $p^2 = m^2$ with residue 1:

$$\Sigma_R(m^2) = 0, \quad \left. \frac{d\Sigma_R}{dp^2} \right|_{p^2=m^2} = 0 \quad (49)$$

Full Propagator

After on-shell renormalization, the renormalized self-energy $\Sigma_R(p^2)$ vanishes at the mass shell, along with its first derivative. The full, exact propagator is:

$$D_R(p^2) = \frac{i}{p^2 - m^2 - \Sigma_R(p^2 + i\epsilon)} \quad (50)$$

Near the mass shell ($p^2 \approx m^2$), the full propagator behaves exactly like the free propagator:

$$D_R(p^2) \xrightarrow{p^2 \rightarrow m^2} \frac{i}{p^2 - m^2 + i\epsilon} \quad (51)$$

Interactions & Correlation Functions

- **1-point:** $\langle \phi \rangle = 0$.
- **3-point (and any odd n-point):** Identically zero ($\langle \Omega | T \phi(x) \phi(y) \phi(z) | \Omega \rangle = 0$) due to exact $\mathbb{Z}_2$ parity.
- **4-point ($2 \rightarrow 2$ scattering):** Given by the 1PI 4-point vertex function $\Gamma^{(4)}(s, t, u)$. To one loop:

$$-i\mathcal{M} = -i\lambda + (-i\lambda)^2 [iV(s) + iV(t) + iV(u)] - i\delta\lambda \quad (52)$$

Where (before renormalization):

$$V(p^2) = \frac{1}{32\pi^2} \int_0^1 dx \left(\frac{2}{\epsilon} - \gamma - \ln \frac{m^2 - x(1-x)p^2}{4\pi} + \mathcal{O}(\epsilon) \right) \quad (53)$$

And using on-shell renormalization conditions

$$\lambda_p = -i\mathcal{M}(p_0^i) \quad p_0^i = (m, \vec{0}) \quad (54)$$

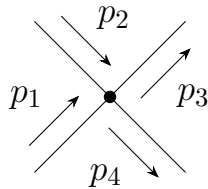
$$s = 4m^2, t = u = 0 \quad (55)$$

Gives:

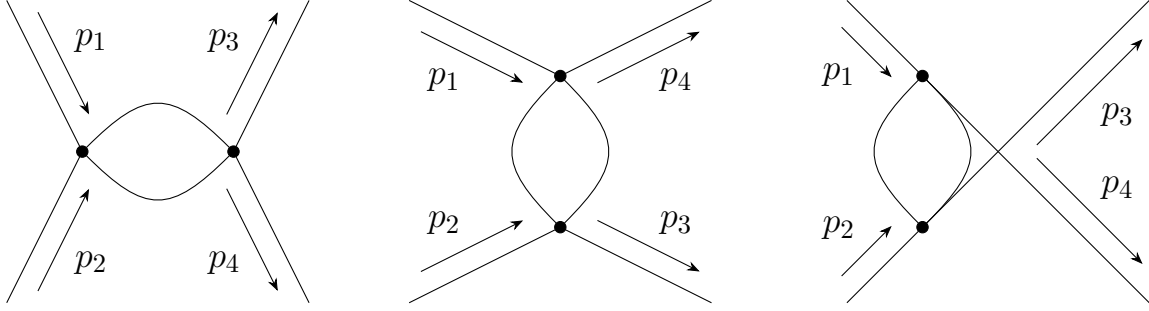
$$\begin{aligned} -\mathcal{M} &= \lambda_p + \lambda_p^2 [V(s) + V(t) + V(u) - V(4m^2) - 2V(0)] = \\ &= \lambda_p + \frac{\lambda_p^2}{32\pi^2} \int_0^1 dx \left[\ln \frac{m^2 - (1-x)s}{m^2 - x(1-x)4m^2} + \ln \frac{m^2 - (1-x)t}{m^2} + \ln \frac{m^2 - (1-x)u}{m^2} \right] \end{aligned} \quad (56)$$

Diagrams

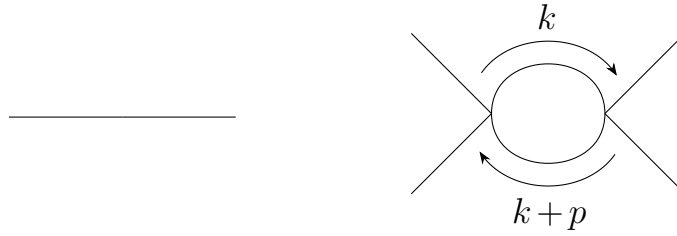
Tree $2 \rightarrow 2$ diagram



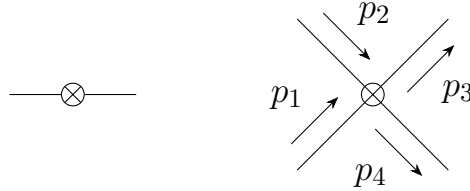
$2 \rightarrow 2$ scattering: S, T, and U channels:



1-loop correction diagrams (Tadpole Self-Energy and Vertex Correction):



Counter terms:



Beta Function

$$\beta(\lambda) = \frac{3\lambda^2}{16\pi^2} \quad (57)$$

4.4 Complex scalar ϕ^4 theory

Full Bare Lagrangian

$$\mathcal{L} = (\partial_\mu \phi^\dagger)(\partial^\mu \phi) - m^2 \phi^\dagger \phi - \frac{\lambda}{4} (\phi^\dagger \phi)^2 \quad (58)$$

Symmetries

Poincaré invariance, Global $U(1)$ charge ($\phi \rightarrow e^{i\alpha} \phi$), Charge Conjugation C ($\phi \leftrightarrow \phi^\dagger$).

Equations of Motion

$$(\square + m^2)\phi + \frac{\lambda}{2}(\phi^\dagger\phi)\phi = 0 \quad (59)$$

Superficial Degree of Divergence

$$D = 4 - N \quad (60)$$

Where N is the number of external legs.

Feynman rules

Element	Position Space	Momentum Space
Directed Propagator	$\Delta_F(x - y)$	$\frac{i}{p^2 - m^2 + i\epsilon}$ (follows arrow)
Vertex (2 in, 2 out)	$-i\lambda \int d^4x$	$-i\lambda(2\pi)^4\delta^{(4)}(\Sigma p)$

Counter terms and Renormalized Lagrangian

$$\mathcal{L}_{CT} = \delta_Z(\partial_\mu\phi^\dagger)(\partial^\mu\phi) - \delta_m\phi^\dagger\phi - \frac{\delta\lambda}{4}(\phi^\dagger\phi)^2 \quad (61)$$

Self-energy counterterm: $i(p^2\delta_Z - \delta_m)$

4-point counterterm: $-i\delta_\lambda$

On-Shell Renormalization

Similar to real ϕ^4 , we demand the pole resides at the physical mass m^2 with residue 1:

$$\Sigma_R(m^2) = 0, \quad \left. \frac{d\Sigma_R}{dp^2} \right|_{p^2=m^2} = 0 \quad (62)$$

And we fix the coupling at threshold: $\lambda_p = -i\mathcal{M}(s = 4m^2, t = 0, u = 0)$.

Full Propagator

After on-shell renormalization, the full exact propagator is:

$$D_R(p^2) = \frac{i}{p^2 - m^2 - \Sigma_R(p^2 + i\epsilon)} \xrightarrow{p^2 \rightarrow m^2} \frac{i}{p^2 - m^2 + i\epsilon} \quad (63)$$

Only the mixed correlator $\langle \Omega | T \phi^\dagger(x) \phi(y) | \Omega \rangle$ is non-zero. $\langle \phi(x) \phi(y) \rangle = 0$ due to charge conservation.

Ward-Takahashi Identity / Noether Current

Global $U(1)$ yields the conserved current $j^\mu = i(\phi^\dagger \partial^\mu \phi - \phi \partial^\mu \phi^\dagger)$ with $\partial_\mu j^\mu = 0$. In correlation functions, this implies the Ward identity:

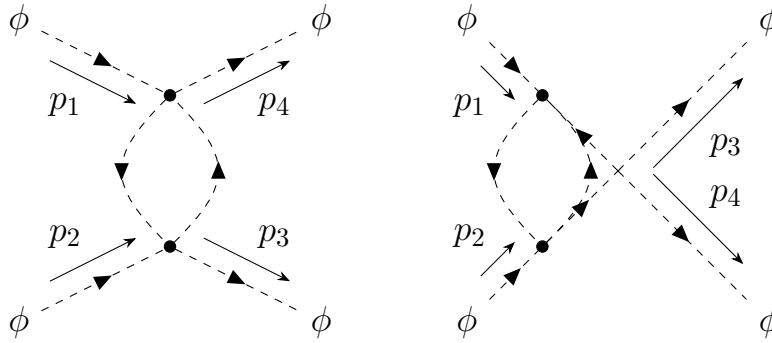
$$\partial_\mu \langle T \{ j^\mu(x) \phi(y_1) \dots \phi^*(z_1) \dots \} \rangle = -i \sum_k \delta^{(4)}(x - y_k) \langle \dots \rangle + i \sum_j \delta^{(4)}(x - z_j) \langle \dots \rangle \quad (64)$$

Interactions & Correlation Functions

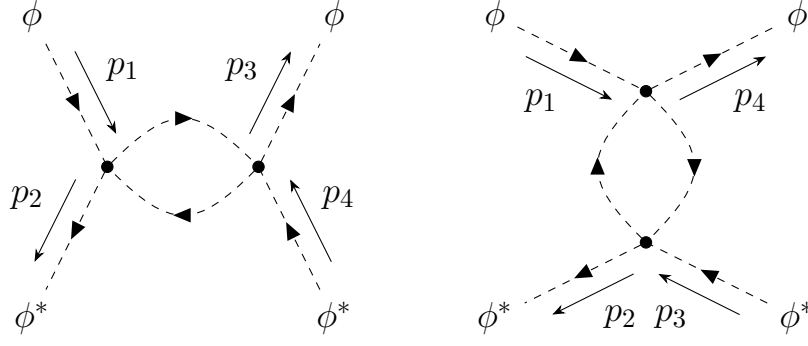
- **1-point:** $\langle \phi \rangle = 0$, by global $U(1)$ charge conservation.
- **3-point:** All 3-point correlation functions (e.g., $\langle \Omega | T \phi^\dagger(x) \phi(y) \phi(z) | \Omega \rangle$) are identically zero because they violate global $U(1)$ charge conservation. The sum of incoming and outgoing charges must be zero.
- **$2 \rightarrow 2$ scattering ($\phi\phi \rightarrow \phi\phi$):** Proceeds only via t and u channels (arrows must align continuously).
- **$2 \rightarrow 2$ scattering ($\phi\phi^* \rightarrow \phi\phi^*$):** Proceeds via s and t channels.

Diagrams

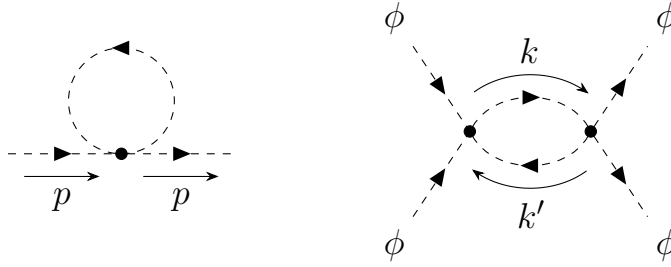
Tree $2 \rightarrow 2$ scattering ($\phi\phi \rightarrow \phi\phi$), T and U channels:



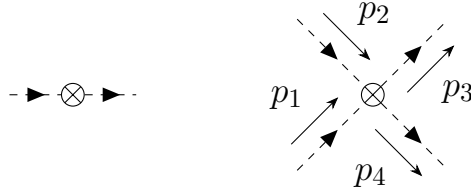
Tree $2 \rightarrow 2$ scattering ($\phi\phi^* \rightarrow \phi\phi^*$), S and T channels:



1-loop diagrams (Self Energy and Vertex Correction):



Counter terms:



Beta Function

$$\beta(\lambda) = \frac{5\lambda^2}{16\pi^2} \quad (65)$$

4.5 Yukawa theory

Full Bare Lagrangian

(Note: ϕ^4 term is *required* for renormalizability because fermion box diagrams generate it).

$$\mathcal{L} = \bar{\psi}(i\not{\partial} - m_f)\psi + \frac{1}{2}(\partial_\mu\phi)^2 - \frac{1}{2}m_s^2\phi^2 - g\bar{\psi}\psi\phi - \frac{\lambda}{4!}\phi^4 \quad (66)$$

Symmetries

Poincaré invariance, Global $U(1)$ (Fermion number). Parity ($\psi \rightarrow \gamma^0 \psi, \phi \rightarrow -\phi$ if pseudoscalar).

Equations of Motion

$$(i\not{\partial} - m_f)\psi = g\phi\psi, \quad (\square + m_s^2)\phi = -g\bar{\psi}\psi - \frac{\lambda}{3!}\phi^3 \quad (67)$$

Superficial Degree of Divergence

$$D = 4 - \frac{3}{2}N_f - N_b \quad (68)$$

Where N_f is the number of external fermions, and N_b is the number of external scalars.

Feynman rules

- Fermion Propagator: $\frac{i(\not{p} + m_f)}{p^2 - m_f^2 + i\epsilon}$
- Scalar Propagator: $\frac{i}{p^2 - m_s^2 + i\epsilon}$
- Yukawa Vertex: $-ig(2\pi)^4 \delta^{(4)}(\Sigma p)$

Counter terms and Renormalized Lagrangian

$$\mathcal{L}_{CT} = \delta_{Z_f} \bar{\psi} i\not{\partial} \psi - \delta_{m_f} \bar{\psi} \psi + \frac{1}{2} \delta_{Z_s} (\partial\phi)^2 - \frac{1}{2} \delta_{m_s} \phi^2 - \delta_g \bar{\psi} \psi \phi - \frac{\delta_\lambda}{4!} \phi^4 + Y\phi \quad (69)$$

Standard insertions for fermion line ($i(\not{p}\delta_{Z_f} - \delta_{m_f})$), scalar line ($i(p^2\delta_{Z_s} - \delta_{m_s})$), vertex ($-i\delta_g$), and the linear tadpole counterterm $Y\phi$.

On-Shell Renormalization

We demand that the exact propagators have poles at their physical masses, with residue 1, and the vertex at zero momentum transfer yields the physical coupling g :

$$\Sigma_{fR}(\not{p} = m_f) = 0, \quad \left. \frac{d\Sigma_{fR}}{d\not{p}} \right|_{\not{p}=m_f} = 0 \quad (70)$$

$$\Sigma_{sR}(p^2 = m_s^2) = 0, \quad \left. \frac{d\Sigma_{sR}}{dp^2} \right|_{p^2=m_s^2} = 0 \quad (71)$$

$$-i\Gamma(p' - p = 0) = -ig \quad (\text{with on-shell fermions } p^2 = p'^2 = m_f^2) \quad (72)$$

Full Propagators

After on-shell renormalization, the full exact propagators are: Fermion:

$$S_{F,R}(p) = \frac{i}{\not{p} - m_f - \Sigma_{fR}(\not{p})} \xrightarrow{p^2 \rightarrow m_f^2} \frac{i}{\not{p} - m_f + i\epsilon} \quad (73)$$

Scalar:

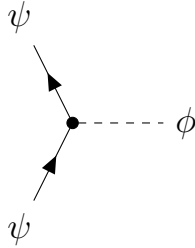
$$D_{F,R}(p) = \frac{i}{p^2 - m_s^2 - \Sigma_{sR}(p^2)} \xrightarrow{p^2 \rightarrow m_s^2} \frac{i}{p^2 - m_s^2 + i\epsilon} \quad (74)$$

Interactions & Correlation Functions

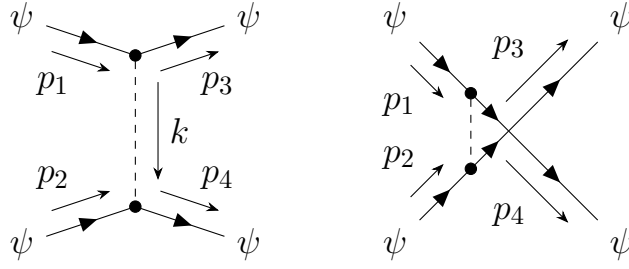
- **1-point:** Fermion loops induce a scalar tadpole $\langle \phi \rangle \neq 0$. The linear counterterm $Y\phi$ is explicitly required to cancel it.
- **3-point:** The fundamental 3-point function is the $\psi\bar{\psi}\phi$ interaction, $\langle \Omega | T\psi(x)\bar{\psi}(y)\phi(z) | \Omega \rangle$, given by the 1PI vertex function $-i\Gamma(p', p)$. At tree level, this is simply $-ig$.
- **2 $\rightarrow$ 2 Fermion scattering ($\psi\psi \rightarrow \psi\psi$):** Proceeds via t and u channel scalar exchange.
- **Effective Potential:** Generates an attractive Yukawa potential $V(r) = -\frac{g^2}{4\pi r}e^{-m_s r}$ in the non-relativistic limit.

Diagrams

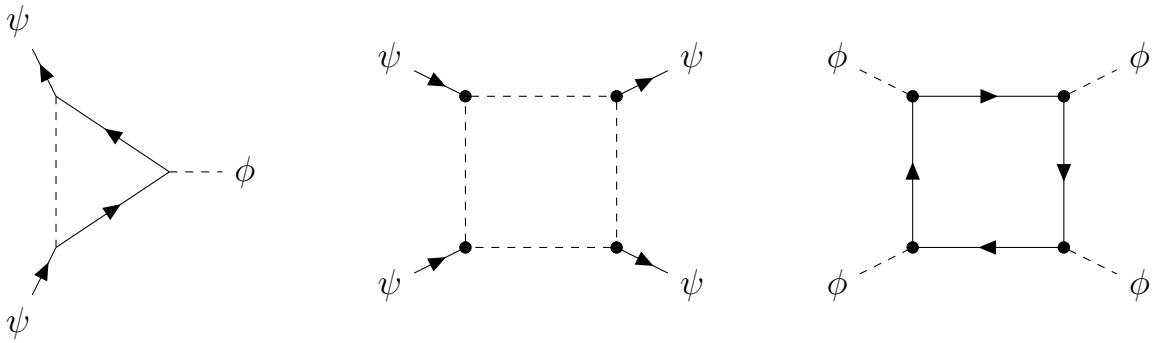
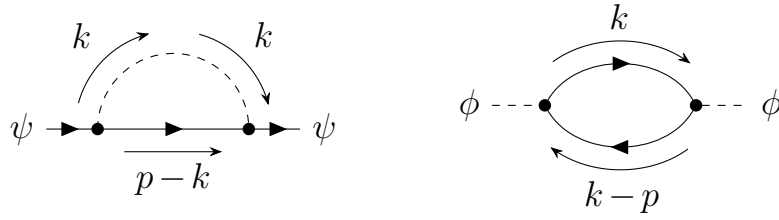
Vertex:



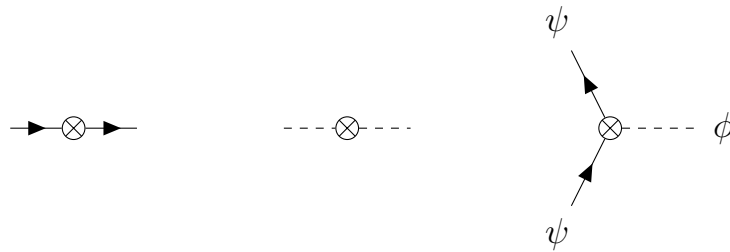
Tree $2 \rightarrow 2$ Fermion scattering ($\psi\psi \rightarrow \psi\psi$), T and U channels:



1-loop diagrams (Self-Energies, Vertex Correction, Box):



Counter terms:



Beta Function

$$\beta(g) = \frac{5g^3}{16\pi^2} \quad (75)$$

4.6 Fermion QED

Full Bare Lagrangian

(Includes Gauge Fixing term)

$$\mathcal{L} = \bar{\psi}(i\not{D} - m)\psi - \frac{1}{4}F_{\mu\nu}F^{\mu\nu} - \frac{1}{2\xi}(\partial_\mu A^\mu)^2, \quad D_\mu = \partial_\mu + ieA_\mu \quad (76)$$

Symmetries

Poincaré invariance, Local $U(1)$ Gauge ($\psi \rightarrow e^{ie\alpha(x)}\psi$, $A_\mu \rightarrow A_\mu - \partial_\mu\alpha$), C, P, T symmetries.

Equations of Motion

$$(i\not{D} - m)\psi = 0, \quad \partial_\mu F^{\mu\nu} + \frac{1}{\xi}\partial^\nu(\partial_\mu A^\mu) = e\bar{\psi}\gamma^\nu\psi \quad (77)$$

Superficial Degree of Divergence

$$D = 4 - \frac{3}{2}N_e - N_\gamma \quad (78)$$

Gauge Fixing

The naive photon propagator cannot be inverted because the kinetic operator $(q^2\eta^{\mu\nu} - q^\mu q^\nu)$ has a zero eigenvalue reflecting gauge redundancy. Using the Faddeev-Popov procedure, we add a gauge-fixing term $-\frac{1}{2\xi}(\partial_\mu A^\mu)^2$, which enforces the Lorentz gauge condition $\partial_\mu A^\mu = 0$ in the path integral. The resulting general bare propagator is:

$$D_F^{\mu\nu}(q) = \frac{-i}{q^2 + i\epsilon} \left(\eta^{\mu\nu} - (1 - \xi) \frac{q^\mu q^\nu}{q^2} \right) \quad (79)$$

Feynman Gauge ($\xi = 1$): Eliminates the $q^\mu q^\nu$ term entirely, yielding $D_F^{\mu\nu}(q) = \frac{-i\eta^{\mu\nu}}{q^2 + i\epsilon}$.

Landau Gauge ($\xi = 0$): Enforces absolute transversality of the propagator, $D_F^{\mu\nu}(q) = \frac{-i}{q^2 + i\epsilon} \left(\eta^{\mu\nu} - \frac{q^\mu q^\nu}{q^2} \right)$.

Feynman rules

- Photon Propagator: $D_F^{\mu\nu}(p)$ (as defined above).
- Fermion Propagator: $\frac{i(\not{p}+m)}{p^2-m^2+i\epsilon}$
- QED Vertex: $-ie\gamma^\mu(2\pi)^4\delta^{(4)}(\Sigma p)$

Counter terms and Renormalized Lagrangian

$$\mathcal{L}_{CT} = i\delta_2\bar{\psi}\not{\partial}\psi - \delta_m\bar{\psi}\psi - \frac{1}{4}\delta_3(F_{\mu\nu})^2 - e\delta_1\bar{\psi}\gamma^\mu A_\mu\psi \quad (80)$$

On-Shell Renormalization

We demand the physical electron mass m , physical field strength, and the macroscopic physical charge e :

$$\Sigma_R(\not{p} = m_e) = 0, \quad \left. \frac{d\Sigma_R}{d\not{p}} \right|_{\not{p}=m_e} = 0 \quad (\text{Fixes } \delta_m, \delta_2) \quad (81)$$

$$\Pi_R(q^2 = 0) = 0 \quad (\text{Fixes } \delta_3, \text{ keeping photon massless}) \quad (82)$$

$$-ie\Gamma^\mu(q = 0) = -ie\gamma^\mu \quad (\text{Fixes } \delta_1 \text{ at threshold scattering limit}) \quad (83)$$

Full Propagators

After on-shell renormalization, the exact propagators are: Fermion:

$$S_{F,R}(p) = \frac{i}{\not{p} - m - \Sigma_R(\not{p})} \xrightarrow{p^2 \rightarrow m^2} \frac{i}{\not{p} - m + i\epsilon} \quad (84)$$

Photon:

$$D_R^{\mu\nu}(q) = \frac{-i}{q^2(1 - \Pi_R(q^2))} \left(\eta^{\mu\nu} - \frac{q^\mu q^\nu}{q^2} \right) - i\xi \frac{q^\mu q^\nu}{q^4} \xrightarrow{q^2 \rightarrow 0} \frac{-i\eta^{\mu\nu}}{q^2} + \text{gauge terms} \quad (85)$$

Ward-Takahashi Identities

The General WT Identity links the 3-point vertex correction to the fermion propagator:

$$q_\mu \Gamma^\mu(p, p+q) = S_{F,R}^{-1}(p+q) - S_{F,R}^{-1}(p) \quad (86)$$

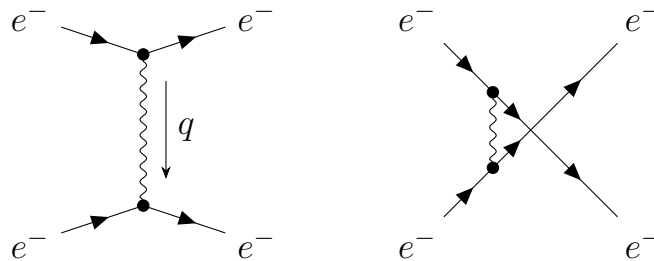
Taking the limit $q \rightarrow 0$ enforces $Z_1 = Z_2$ (and thus $\delta_1 = \delta_2$). This ensures the physical charge e is renormalized identically for all species uniquely by the vacuum polarization Z_3 . Furthermore, gauge invariance mandates the transversality of the Photon Self-Energy: $q_\mu \Pi^{\mu\nu}(q) = 0$.

Interactions & Correlation Functions

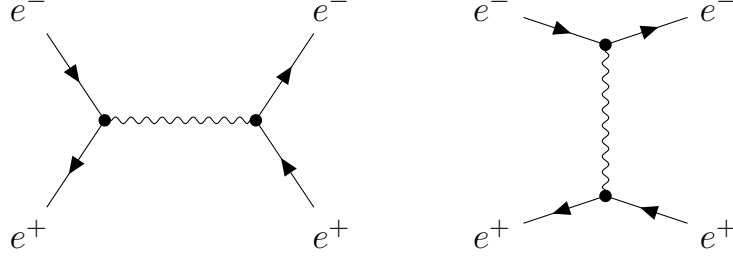
- **3-point (Furry's Theorem):** The 3-photon vertex $\langle \Omega | T A_\mu(x) A_\nu(y) A_\rho(z) | \Omega \rangle = 0$ is identically zero due to Charge Conjugation symmetry. Any closed fermion loop with an odd number of photon vertices vanishes.
- **3-point (Vertex):** The fermion-photon vertex $\langle \psi \bar{\psi} A_\mu \rangle$ yields the dressed vertex $\Gamma^\mu(p', p) = \gamma^\mu F_1(q^2) + \frac{i\sigma^{\mu\nu} q_\nu}{2m} F_2(q^2)$. Anomalous magnetic moment $a_e = \frac{g-2}{2} = F_2(0) = \frac{\alpha}{2\pi}$.
- **Vacuum Polarization:** $\Pi^{\mu\nu}(q) = (q^2 \eta^{\mu\nu} - q^\mu q^\nu) \Pi(q^2)$.
- **Møller Scattering** ($e^- e^- \rightarrow e^- e^-$): t and u channel photon exchange.
- **Bhabha Scattering** ($e^- e^+ \rightarrow e^- e^+$): s and t channel photon exchange.
- **Compton Scattering** ($e^- \gamma \rightarrow e^- \gamma$): s and u channel fermion exchange.

Diagrams

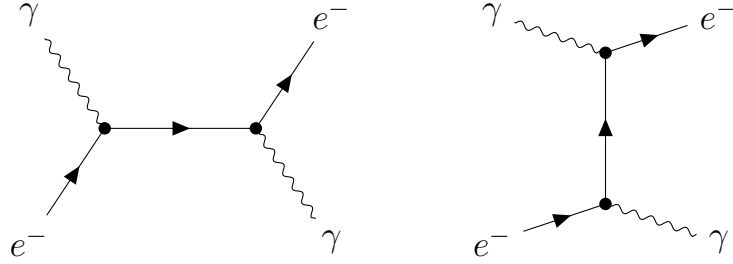
Tree $2 \rightarrow 2$ scattering (Møller $e^- e^- \rightarrow e^- e^-$), T and U channels:



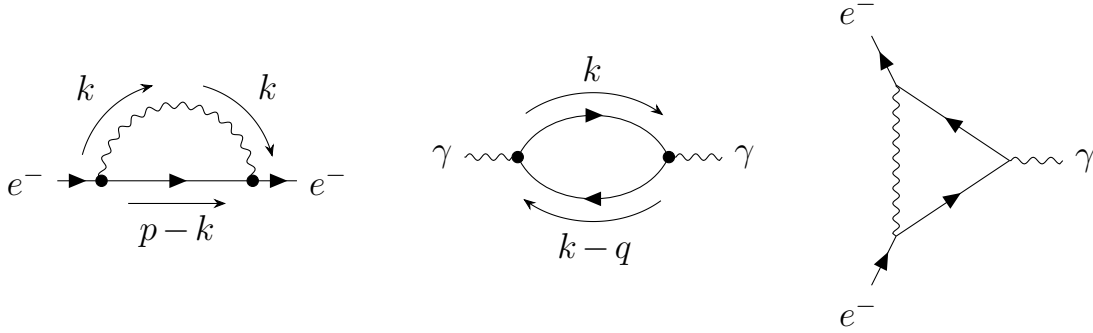
Tree $2 \rightarrow 2$ scattering (Bhabha $e^- e^+ \rightarrow e^- e^+$), S and T channels:



Tree $2 \rightarrow 2$ scattering (Compton $e^- \gamma \rightarrow e^- \gamma$), S and U channels:



1-loop diagrams (Electron Self-Energy, Vacuum Polarization, Vertex Correction):



Beta Function

$$\beta(e) = \frac{e^3}{12\pi^2} \quad (87)$$

IR Divergences and Bremsstrahlung

When an electrically charged particle undergoes scattering, the probability of emitting a very low-energy (soft) photon (Bremsstrahlung) diverges. The cross-section for emitting a photon with energy less than some detector resolution E_{cut} diverges logarithmically at the lower bound (as $\mu_{IR} \rightarrow 0$):

$$\sigma_{\text{Brem}} \propto \alpha \int_{\mu_{IR}}^{E_{\text{cut}}} \frac{dE_{\gamma}}{E_{\gamma}} \sim \alpha \ln \left(\frac{E_{\text{cut}}}{\mu_{IR}} \right) \quad (88)$$

Simultaneously, the 1-loop virtual vertex correction $F_1(q^2)$ generates a negative logarithmic divergence of the exact same form due to soft virtual photons in the loop.

The Bloch-Nordsieck Theorem: Any physical detector has a finite energy resolution E_{cut} , meaning it cannot distinguish between a pure elastic scattering event and an inelastic scattering event where an unobservably soft photon is emitted. The physically measurable cross section is the sum:

$$\sigma_{\text{obs}} = \sigma_{\text{virtual}} + \sigma_{\text{Brem}} \quad (89)$$

In this sum, the IR divergence $\ln(\mu_{IR})$ perfectly cancels out to all orders in perturbation theory, leaving a finite, observable result that depends only on the detector's cutoff E_{cut} .

4.7 Scalar QED

Full Bare Lagrangian

(Includes quartic $(\phi^\dagger\phi)^2$ term required for renormalization, and gauge fixing).

$$\mathcal{L} = (D_\mu\phi)^\dagger(D^\mu\phi) - m^2\phi^\dagger\phi - \frac{\lambda}{4}(\phi^\dagger\phi)^2 - \frac{1}{4}F_{\mu\nu}F^{\mu\nu} - \frac{1}{2\xi}(\partial_\mu A^\mu)^2 \quad (90)$$

Symmetries

Poincaré invariance, Local $U(1)$ Gauge ($\phi \rightarrow e^{ie\alpha(x)}\phi$, $A_\mu \rightarrow A_\mu - \partial_\mu\alpha$), C, P, T symmetries.

Equations of Motion

$$D_\mu D^\mu\phi + m^2\phi + \frac{\lambda}{2}(\phi^\dagger\phi)\phi = 0, \quad \partial_\mu F^{\mu\nu} + \frac{1}{\xi}\partial^\nu(\partial_\mu A^\mu) = ie(\phi^\dagger D^\nu\phi - (D^\nu\phi^\dagger)\phi) \quad (91)$$

Superficial Degree of Divergence

$$D = 4 - N_\phi - N_\gamma \quad (92)$$

Feynman rules

- Cubic Vertex ($A_\mu \phi^\dagger \phi$): $-ie(p+p')^\mu$ (where p, p' are incoming/outgoing scalar momenta)
- Quartic (Seagull) Vertex ($A_\mu A_\nu \phi^\dagger \phi$): $2ie^2 \eta^{\mu\nu}$

Counter terms and Renormalized Lagrangian

$$\mathcal{L}_{CT} = \delta_Z |D_\mu \phi|^2 - \delta_m |\phi|^2 - \frac{\delta_\lambda}{4} |\phi|^4 - \frac{1}{4} \delta_A (F_{\mu\nu})^2 \quad (93)$$

By gauge invariance, the counterterms for the cubic vertex ($-ie\delta_1(p+p')^\mu$) and quartic vertex ($2ie^2\delta_2\eta^{\mu\nu}$) are strictly fixed to match the field renormalization: $\delta_1 = \delta_2 = \delta_Z$.

On-Shell Renormalization

We apply identical on-shell conditions as in Fermion QED to fix the scalar mass m , maintain a massless photon, and fix the charge e :

$$\Sigma_R(m^2) = 0, \quad \left. \frac{d\Sigma_R}{dp^2} \right|_{p^2=m^2} = 0, \quad \Pi_R(0) = 0 \quad (94)$$

Full Propagators

After on-shell renormalization, the exact propagators are: Scalar:

$$D_{F,R}(p) = \frac{i}{p^2 - m^2 - \Sigma_R(p^2)} \xrightarrow{p^2 \rightarrow m^2} \frac{i}{p^2 - m^2 + i\epsilon} \quad (95)$$

Photon: (Identical to Fermion QED).

Ward-Takahashi Identities

Analogous to Fermion QED, ensuring charge conservation and transversality of the photon self-energy tensor $\Pi^{\mu\nu}(q)$.

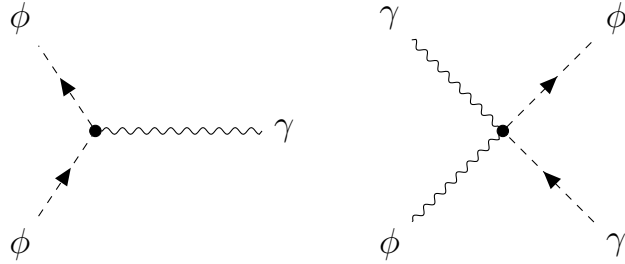
$$q_\mu \Gamma^\mu(p, p+q) = \Delta_{F,R}^{-1}(p+q) - \Delta_{F,R}^{-1}(p) \quad (96)$$

Interactions & Correlation Functions

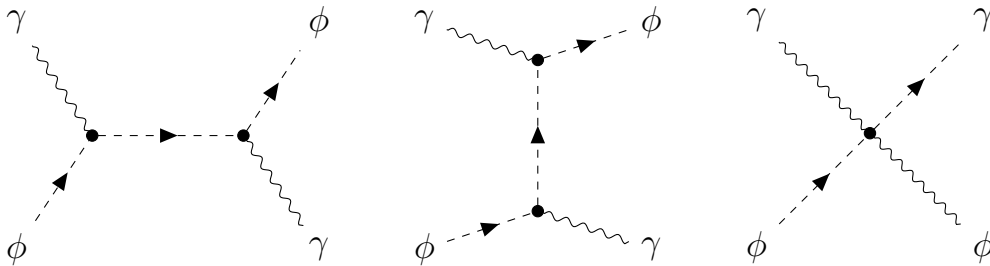
- **1-point:** Gauge invariance prohibits scalar tadpoles $\langle\phi\rangle = 0$.
- **3-point (Furry's Theorem analog):** The 3-photon vertex $\langle A_\mu A_\nu A_\rho\rangle = 0$ by Charge Conjugation symmetry.
- **3-point (Vertex):** The fundamental 3-point function is the scalar-photon vertex $\langle\phi^\dagger\phi A_\mu\rangle$, yielding the dressed vertex $\Gamma^\mu(p',p)$.
- **$2 \rightarrow 2$ Scattering:** Compton scattering $\gamma\phi \rightarrow \gamma\phi$ proceeds via s and u channels (cubic vertex) PLUS a contact term from the quartic seagull vertex. This seagull diagram is strictly necessary to satisfy gauge invariance and the Ward Identity.

Diagrams

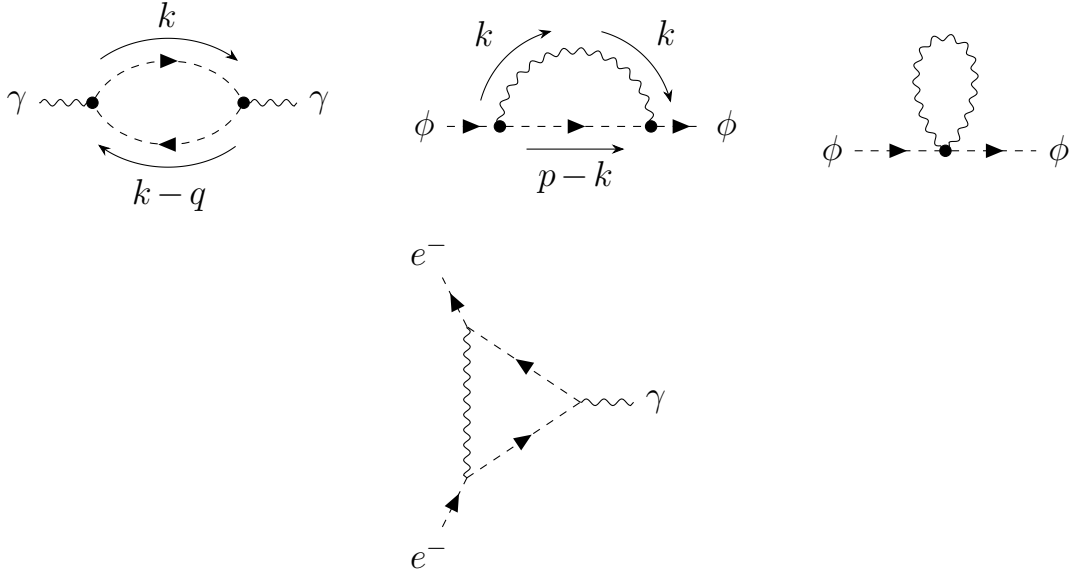
Vertices:



Tree $2 \rightarrow 2$ scattering (Compton $\gamma\phi \rightarrow \gamma\phi$), S, U, and Seagull channels:



1-loop diagrams (Vacuum Polarization, Scalar Self-Energy Photon Loop, Seagull Tadpole):



Beta Function

$$\beta(e) = \frac{e^3}{48\pi^2} \quad (97)$$

(The scalar loop contribution to vacuum polarization is 1/4 that of a Dirac fermion).

5 Math

5.1 Gaussian Integrals

Simple Gaussian:

$$\int dx e^{-ax^2+bx} = \sqrt{\frac{\pi}{a}} e^{\frac{b^2}{4a}} \quad (98)$$

N -dimensional, with source J :

$$\int d^n x e^{-\frac{1}{2}x^T A x + J^T x} = \sqrt{\frac{(2\pi)^N}{\det A}} \exp\left(\frac{1}{2}J^T A^{-1}J\right) \quad (99)$$

Grassmann Numbers (assuming standard anticommutation $\theta_i\theta_j = -\theta_j\theta_i$):

$$\int \left(\prod_{i=1}^N d\bar{\theta}_i d\theta_i \right) \exp(-\bar{\theta}M\theta + \bar{\eta}\theta + \bar{\theta}\eta) = (\det M) \exp(\bar{\eta}M^{-1}\eta) \quad (100)$$

Functional Gaussian (fields path integral in continuous spacetime):

$$\begin{aligned} \int \mathcal{D}\phi \exp\left(-\frac{1}{2} \int d^4x d^4y \phi(x) A(x,y) \phi(y) + \int d^4x J(x) \phi(x)\right) &\propto \\ &\propto (\det A)^{-1/2} \exp\left(\frac{1}{2} \int d^4x d^4y J(x) A^{-1}(x,y) J(y)\right) \end{aligned} \quad (101)$$

5.2 Feynman's trick

Two variables:

$$\frac{1}{AB} = \int_0^1 dx \int_0^1 dy \frac{\delta(x+y-1)}{[xA+yB]^2} = \int_0^1 dx \frac{1}{[xA+(1-x)B]^2} \quad (102)$$

Three variables:

$$\begin{aligned} \frac{1}{ABC} &= 2 \int_0^1 dx \int_0^1 dy \int_0^1 dz \frac{\delta(x+y+z-1)}{[xA+yB+zC]^3} = \\ &= 2 \int_0^1 dx \int_0^{1-x} dy \frac{1}{[xA+yB+(1-x-y)C]^3} \end{aligned} \quad (103)$$

General:

$$\frac{1}{A_1 \dots A_n} = (n-1)! \int_0^1 dx_1 \dots dx_n \delta(1 - \sum x_i) \frac{1}{(x_1 A_1 + \dots + x_n A_n)^n} \quad (104)$$

5.3 Grassman Numbers

Definition: A set of elements θ_i that completely anticommute: $\{\theta_i, \theta_j\} = \theta_i\theta_j + \theta_j\theta_i = 0$. General term in the Grassman algebra of n Grassman numbers θ_i : Because $\theta_i^2 = 0$, any function has a finite Taylor series. For 2 variables: $f(\theta_1, \theta_2) = A + B\theta_1 + C\theta_2 + D\theta_1\theta_2$.

Properties:

- Nilpotency: $\theta_i^2 = 0$.
- Commute with normal c-numbers, anticommute with other Grassmanns.

Some basic functions:

$$e^\theta = 1 + \theta \quad (105)$$

$$\cos \theta = 1, \quad \sin \theta = \theta \quad (106)$$

$$\cosh \theta = 1, \quad \sinh \theta = \theta \quad (107)$$

Differentiation and Integration: Left differentiation: $\frac{\partial}{\partial \theta_i}(\theta_i\theta_j) = \theta_j$. Integration rules (Berezin integral):

$$\int d\theta = 0, \quad \int \theta d\theta = 1 \quad (108)$$

Delta function: $\delta(\theta) = \theta$. Complex Grassman numbers: Formed by linear combinations: $\theta = \frac{1}{\sqrt{2}}(\theta_1 + i\theta_2)$, $\bar{\theta} = \frac{1}{\sqrt{2}}(\theta_1 - i\theta_2)$. Real Grassmann variables describe neutral fermions (Majorana), while complex describe charged fermions (Dirac). Conjugation reverses order: $(\theta\eta)^* = \eta^*\theta^*$. Gaussian Integrals:

$$\int d\bar{\theta}d\theta e^{-a\bar{\theta}\theta} = \int d\bar{\theta}d\theta (1 - a\bar{\theta}\theta) = a \quad (109)$$

5.4 Pauli matrices

Definition:

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \quad (110)$$

Properties:

$$[\sigma_i, \sigma_j] = 2i\epsilon_{ijk}\sigma_k, \quad \{\sigma_i, \sigma_j\} = 2\delta_{ij}\mathbb{I}, \quad \text{Tr}(\sigma_i) = 0 \quad (111)$$

5.5 Gamma matrices

Matrices satisfying the Dirac/Clifford algebra: $\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}\mathbb{1}_{4\times 4}$. Properties:

$$\text{Tr}(\mathbb{1}) = 4 \quad (112)$$

$$\text{Tr}(\text{any odd number of } \gamma) = 0 \quad (113)$$

$$\text{Tr}(\gamma^\mu \gamma^\nu) = 4\eta^{\mu\nu} \quad (114)$$

$$\text{Tr}(\gamma^\mu \gamma^\nu \gamma^\rho \gamma^\sigma) = 4(\eta^{\mu\nu}\eta^{\rho\sigma} - \eta^{\mu\rho}\eta^{\nu\sigma} + \eta^{\mu\sigma}\eta^{\nu\rho}) \quad (115)$$

$$\gamma^\mu \gamma_\mu = 4 \quad (116)$$

$$\gamma^\mu \not{a} \gamma_\mu = -2\not{a} \quad (117)$$

5.6 Inverting Fourier series

If $f(x) = \sum_k e^{ikx} c_k$, then $c_k = \frac{1}{L} \int_0^L f(x) e^{-ikx} dx$.

5.7 Fourier transform

$$f(x) = \int \frac{d^4 k}{(2\pi)^4} e^{-ik \cdot x} \tilde{f}(k), \quad \tilde{f}(k) = \int d^4 x e^{ik \cdot x} f(x) \quad (118)$$

5.8 Principal value

Used in contour integration for resolving poles on the real axis:

$$\lim_{\epsilon \rightarrow 0} \frac{1}{x \pm i\epsilon} = \mathcal{P}\left(\frac{1}{x}\right) \mp i\pi\delta(x) \quad (119)$$

5.9 Gamma function

Definition:

$$\Gamma(z) = \int_0^\infty x^{z-1} e^{-x} dx \quad (\text{for } \text{Re}(z) > 0) \quad (120)$$

Properties:

$$\Gamma(z+1) = z\Gamma(z) \quad (121)$$

$$\Gamma(n) = (n-1)! \quad (\text{for integer } n) \quad (122)$$

$$\Gamma(1/2) = \sqrt{\pi} \quad (123)$$

It has simple poles at $z = 0, -1, -2, \dots$ with residue $\frac{(-1)^n}{n!}$.

5.10 The Residue Theorem

If $f(z)$ is analytic inside and on some closed contour C except at a finite number of isolated singularities a_k inside C :

$$\oint_C f(z)dz = 2\pi i \sum_{k=1}^n \text{Res}(f, a_k) \quad (124)$$

where for a simple pole, the residue is $\text{Res}(f, c) = \lim_{z \rightarrow c} (z - c)f(z)$.